

1. Python Fundamentals (Beginner to Advanced)

- Python features: Interpreted, Dynamic Typing, OOP, Large Libraries.
- Data Types: int, float, str, list, tuple, set, dict, bool.
- Mutable vs Immutable types.
- List vs Tuple vs Set vs Dictionary differences.
- Type Casting and Type Checking (type(), isinstance()).
- Memory Management and Garbage Collection.
- Deep Copy vs Shallow Copy.
- PEP 8 Coding Standards.
- Virtual Environment (venv) usage.

2. Object Oriented Programming

- Class and Object concepts.
- Constructor (`__init__`) and dunder methods.
- Encapsulation (private variables using `__`).
- Abstraction (using ABC module).
- Inheritance types (Single, Multiple, Multilevel).
- Polymorphism and Method Overriding.
- `super()` usage.
- Composition vs Inheritance.

3. Advanced Python Concepts

- Lambda Functions.
- List/Dict/Set Comprehensions.
- Generators and yield keyword.
- Decorators (function & class based).
- Iterators and `__iter__`, `__next__`.
- Context Managers and with statement.
- Multithreading vs Multiprocessing.
- Global Interpreter Lock (GIL).
- Asyncio basics.

4. Data Structures & Algorithms (DSA)

- Time Complexity ($O(1)$, $O(n)$, $O(\log n)$, $O(n^2)$).
- Arrays and Strings common problems.
- Linked List implementation.
- Stack and Queue (using list & collections.deque).
- Recursion and Backtracking.
- Sorting Algorithms (Bubble, Merge, Quick).
- Searching Algorithms (Binary Search).
- Hashing and Dictionary usage.
- Two Pointer Technique.
- Sliding Window Technique.

5. Exception Handling & File Handling

- try, except, else, finally usage.
- Custom Exceptions.
- Reading/Writing files using with statement.
- Handling CSV and JSON files.
- Working with large files efficiently.

6. Flask Web Development

- What is Flask? Micro web framework.
- Project Structure best practices.
- Routing using `@app.route`.
- Templates using Jinja2.
- Forms handling (GET vs POST).
- Flask with SQLAlchemy ORM.
- Blueprints for large apps.
- REST API creation in Flask.
- Authentication basics.
- Deploying Flask app.

7. Django Basics (Interview Ready)

- MTV Architecture (Model-Template-View).
- Django project vs app.
- Models and Migrations.
- Admin Panel usage.
- Django ORM queries.
- Forms and ModelForms.
- Authentication system.
- Django REST Framework basics.

8. Database & SQL

- Primary Key vs Foreign Key.
- Indexes and Constraints.
- JOIN types (INNER, LEFT, RIGHT).
- Normalization basics.
- Transactions and ACID properties.

9. Git & Deployment

- git init, add, commit, push, pull.
- Branching and merging.
- Resolving merge conflicts.
- Deploying to VPS/Cloud.
- Environment variables & security.

10. HR & Behavioral Interview Questions

- Tell me about yourself (structure answer).
- Explain one project deeply.
- Challenges faced in project.
- Strengths and weaknesses.
- Why should we hire you?
- Future goals as Python developer.